Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________

Mixed Practice: Vertex Form and Graphing

Example (Mixed Practice: Vertex Form and Graphing):
For $y = 2(x-1)^2 - 3$: vertex $(1, -3)$, opens up; y-intercept $y = 2(0-1)^2 - 3 = -1$, point $(0, -1)$, mirror point $(2, -1)$.

Directions: For each function, find the vertex, direction of opening, y-intercept, and its mirror point.

1. $y = (x+2)^2 - 1$

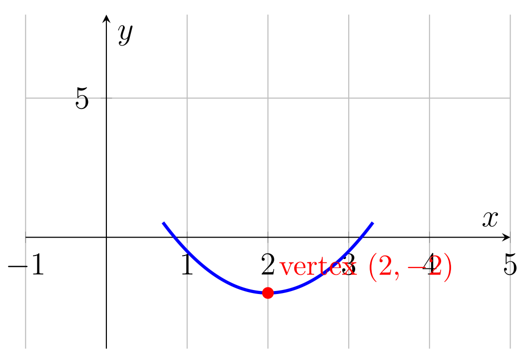
2. $y = -3(x-1)^2 + 4$

3. $y = \frac{1}{2}(x-4)^2 + 2$

4. $y = -(x+3)^2 + 2$

Mixed Practice: Transformations and Max/Min

Example (Mixed Practice: Transformations and Max/Min):
For $y = -(x+4)^2 + 7$: transformations are shift left 4, reflect over x-axis, shift up 7; since $a = -1 < 0$, maximum value is 7 at $x = -4$.



Reference graph of $y = 1.5(x-2)^2 - 2$.

Directions: Describe the transformations from $y = x^2$ and state the max/min value for each function.

5. $y = 1.5(x-2)^2 - 2$ (use the graph above to confirm the vertex)

6. $y = -2(x+1)^2 + 5$

7. $y = 3(x-2)^2 - 1$

8. A ball's height is $h(t) = -16t^2 + 32t + 4$. Find the maximum height and when it occurs.

Challenge (same sheet):

9. Create your own vertex-form equation with a maximum value of 10 at $x = -2$, and describe all its transformations from $y = x^2$.

Answer Key — fully worked

Procedure: Identify a, h, and k from the equation.; State the vertex, direction of opening, and axis of symmetry.; Find the y-intercept by substituting $x = 0$, and its mirror point across the axis of symmetry.; Describe the transformations from $y = x^2$: shifts, reflection, stretch/compression.; State the max/min value (k) and where it occurs ($x = h$), interpreting units for word problems..

1. Vertex $(-2, -1)$, opens up; y-intercept $y = (0 + 2)^2 - 1 = 3$, point $(0, 3)$, mirror point $(-4, 3)$.

2. Vertex $(1, 4)$, opens down; y-intercept $y = -3(0-1)^2+4 = 1$, point $(0, 1)$, mirror point $(2, 1)$.

3. Vertex $(4, 2)$, opens up; y-intercept $y = \frac{1}{2}(0 - 4)^2 + 2 = 10$, point $(0, 10)$, mirror point $(8, 10)$.

4. Vertex $(-3, 2)$, opens down; y-intercept $y = -(0 + 3)^2 + 2 = -7$, point $(0, -7)$, mirror point $(-6, -7)$.

5. Shift right 2, stretch narrower by 1.5, shift down 2; vertex $(2, -2)$, minimum value -2 – matches the graph shown.

6. Shift left 1, reflect over x-axis, stretch narrower by 2, shift up 5; maximum value 5 at $x = -1$.

7. Shift right 2, stretch narrower by 3, shift down 1; vertex $(2, -1)$, minimum value -1.

8. $t = -\frac{32}{-32} = 1$; $h(1) = -16 + 32 + 4 = 20$. Maximum height is 20 feet, occurring at 1 second.

9. A maximum value of 10 at $x = -2$ means the vertex is $(-2, 10)$ with $a < 0$, so $h = -2$, $k = 10$. One valid answer: $y = -(x + 2)^2 + 10$. Transformations from $y = x^2$: shift left 2, reflect over the x-axis, shift up 10 (any $a < 0$ works, e.g. $y = -3(x + 2)^2 + 10$ would also add a narrowing stretch by 3).